

PKCERT AI & Software Development Internship

Task 01 — Development Environment Setup & GitHub Fundamentals

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Objective

The objective of this task was to configure a complete Python development environment, install and set up Visual Studio Code with the required extensions, create and activate a virtual environment, and understand the fundamentals of GitHub and version control by creating a repository, making a first commit, pushing changes, and working with branches.

Part A — Development Environment Setup

For the environment setup I installed Python (version 3.11 or above) and Visual Studio Code, then added the required VS Code extensions: **Python**, **Pylance**, **Jupyter**, and **GitLens**. Finally, I created and activated a Python virtual environment for the project.

Figures [1](#) and [2](#) show the installed VS Code extensions, including GitLens, the Jupyter extension family, Pylance, and the core Python extension.

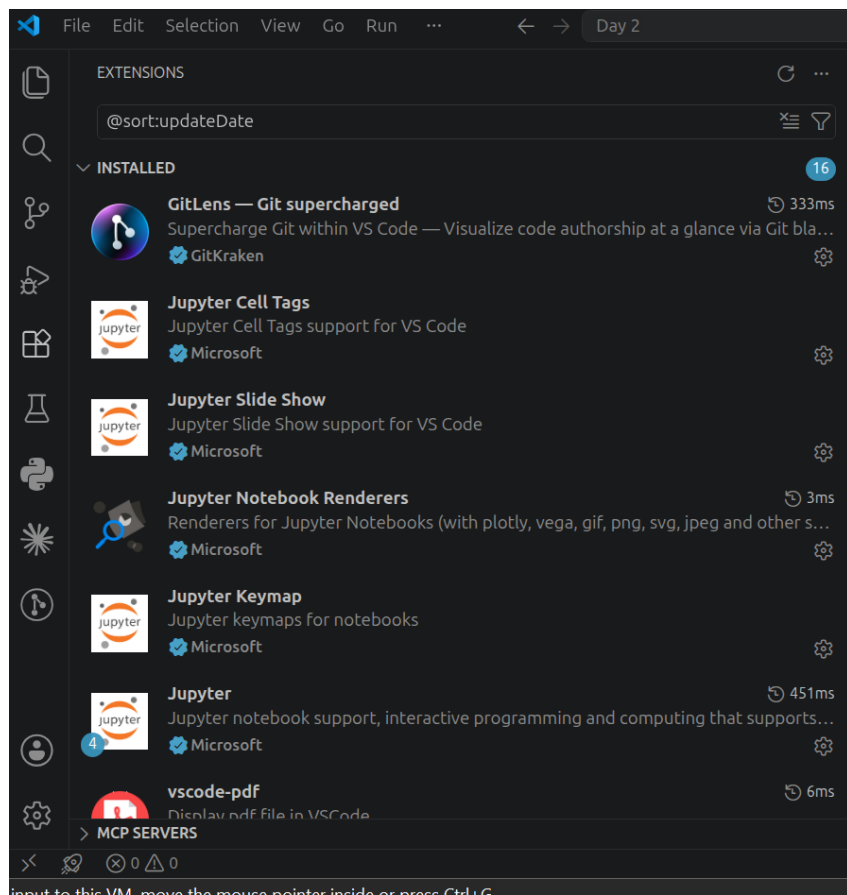


Figure 1: VS Code Extensions view showing GitLens and the Jupyter extensions installed.

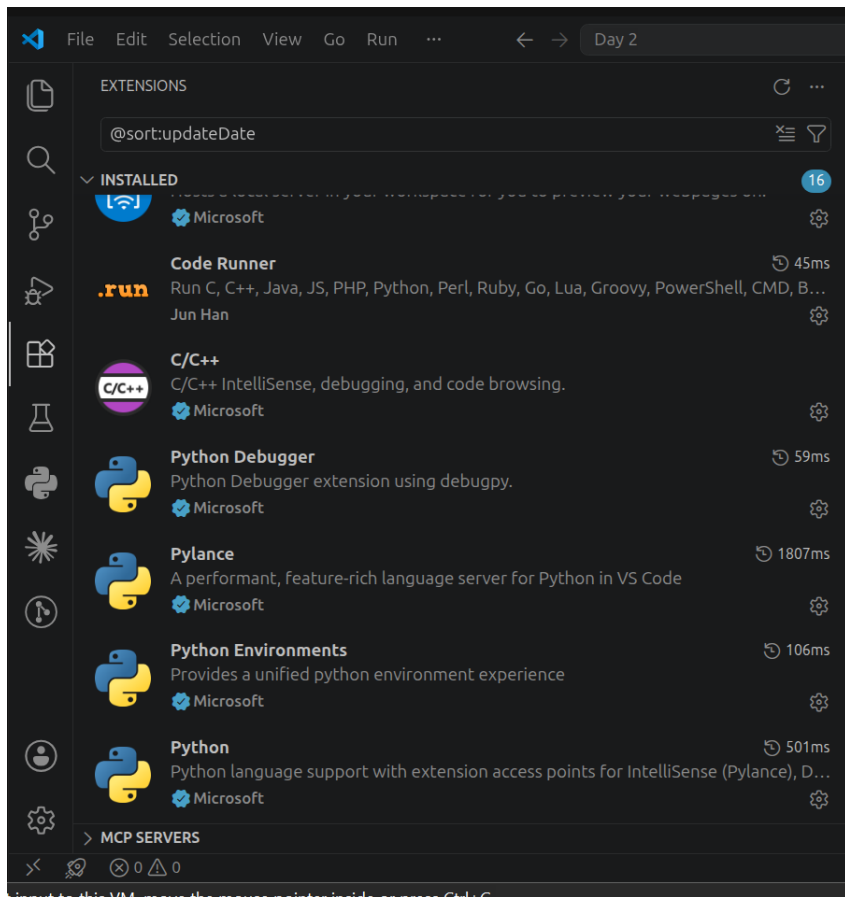


Figure 2: VS Code Extensions view showing Pylance, the Python extension, Python Debugger, and other supporting tools.

The screenshot in Figure 3 confirms the active virtual environment (`venv_build`) and the installed Python version reported by `python3 --version` as **Python 3.14.4**, which satisfies the “3.11 or above” requirement.

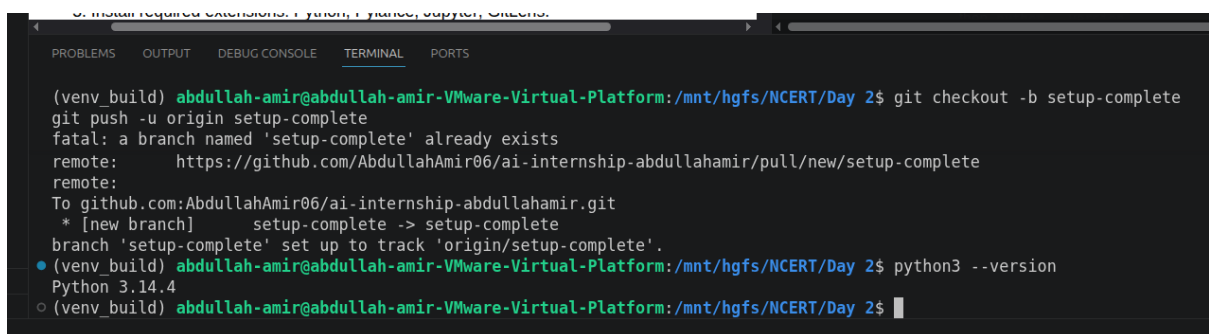


Figure 3: Integrated terminal showing the active `venv_build` virtual environment, the `setup-complete` branch being pushed, and the verified Python 3.14.4 installation.

Part B — GitHub Repository Setup

Using my existing GitHub account, I created a public repository following the required naming format: `ai-internship-abdullahamir`. I then cloned the repository onto my local machine so

I could add project files and manage version control locally.

Repository link: <https://github.com/AbdullahAmir06/ai-internship-abdullahamir>

Part C — Git Fundamentals

For the Git fundamentals portion I completed the following steps:

- Created a `README.md` file containing my name, batch, and domain.
- Added the project files (`README.md`, `welcome.py`, and `.gitignore`) to the repository.
- Made my first commit with the meaningful message: *“Initial commit: environment setup and bonus welcome script”*.
- Pushed the changes to GitHub on the `main` branch.
- Created a new branch named `setup-complete` and pushed it to the remote (as shown earlier in Figure 3).

Figure 4 shows the final state of the repository on GitHub, including the `main` branch, the two branches, the single initial commit, the committed files, and the rendered README.

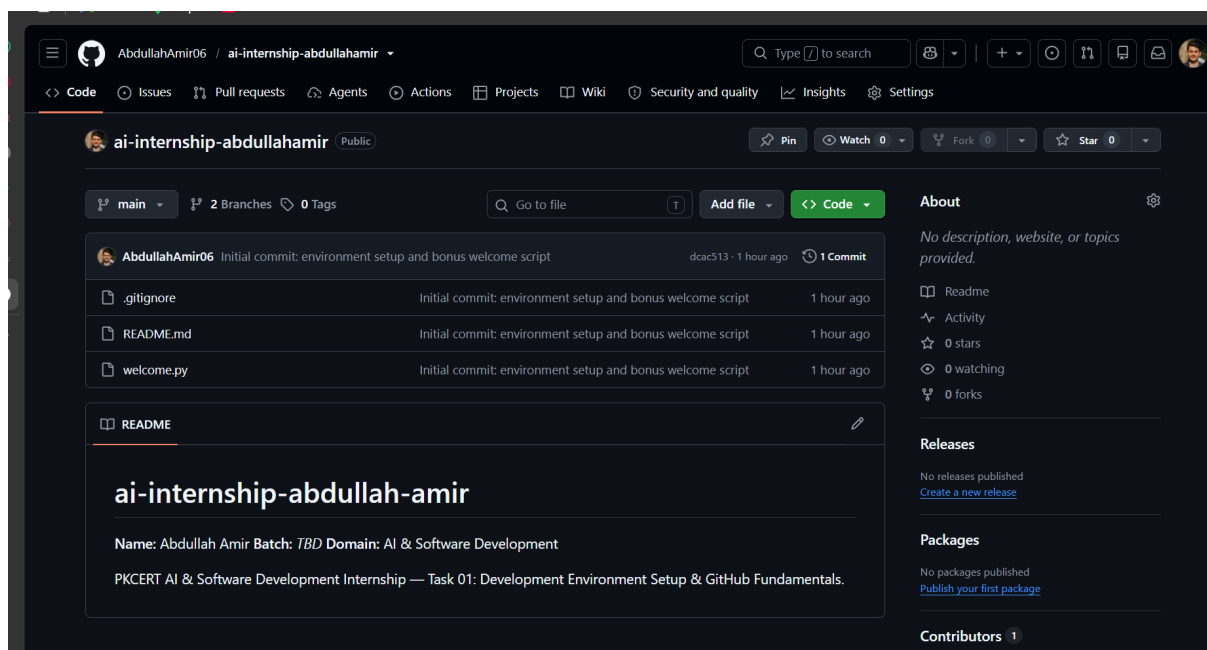


Figure 4: The GitHub repository `ai-internship-abdullahamir` showing the pushed files, commit history, two branches, and the rendered README.

Bonus Task — Welcome Program

For the bonus task, I wrote a short Python program (`welcome.py`) that asks the user for their name and prints a personalized welcome message:

```
name = input("Enter your name: ")
print(f"Welcome, {name}!")
```

Conclusion

Completing this task gave me hands-on experience setting up a professional Python development environment with Visual Studio Code and its essential extensions, and working with an isolated virtual environment. More importantly, it strengthened my understanding of Git and GitHub fundamentals — creating a repository, committing with meaningful messages, pushing changes, and managing branches — which form the foundation for collaborative software development throughout the rest of the internship.